Market Calibration of the CGMY Process to SPX Implied-Volatility Surfaces

Section 7: Methodology, Implementation, and Results

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Contents

1 Overview

This report documents the end-to-end market calibration of the Carr–Géman–Madan–Yor (CGMY) tempered stable process to S&P 500 (SPX) implied-volatility (IV) surfaces observed on three snapshot dates spanning markedly different market regimes:

- **2020-01-15** — low-volatility, pre-COVID environment (VIX $\approx$ 13).
- **2020-03-18** — acute COVID-19 crisis (VIX $\approx$ 76, largest single-day SPX declines on record).
- **2021-09-15** — post-COVID recovery (VIX $\approx$ 20).

The calibration exploits the B-Spline Galerkin + Crank–Nicolson (CN) solver developed in the preceding sections of this work as its inner pricing engine. No external option-pricing library is used. Optimisation is performed with a two-phase scheme (Differential Evolution followed by L-BFGS-B) over an unconstrained reparametrisation of the CGMY parameter space.

2 The CGMY Model

The CGMY Lévy process has characteristic exponent

$$\Psi(u) = C\,\Gamma(-Y)\big[(M-iu)^Y - M^Y + (G+iu)^Y - G^Y\big], \tag{1}$$

where $C > 0$ (overall jump intensity), $G > 0$ (left-tail decay), $M > 1$ (right-tail decay), $Y \in (0,2)\setminus\{1\}$ (fine-structure index). For $Y \in (1,2)$ the process has infinite variation and is the dominant regime for equity markets.

Under the risk-neutral measure the log-price dynamics are

$$\log(S_T/S_0) = (r - q - \omega_S)\,T + X_T,$$

where X_T is a centred CGMY process and $\omega_S = C\,\Gamma(-Y)\big[(M-1)^Y - M^Y + (G+1)^Y - G^Y\big]$ is the martingale correction.

3 Data and Surface Construction

3.1 Raw data

Raw option chains are sourced from OptionMetrics for three snapshot dates. Each chain contains end-of-day bid, ask, and implied volatility for all listed SPX calls and puts.

3.2 Quality filtering (Step 2)

The following filters are applied to each raw chain before any further processing:

1. Remove quotes with zero bid, zero open interest, or non-positive mid-price.
2. Remove quotes with negative implied volatility.
3. Remove quotes with bid $>$ ask (crossed markets).
4. Compute mid-price $= \frac{1}{2}(\text{bid} + \text{ask})$ and relative spread $= (\text{ask} - \text{bid})/\text{mid}$.

Two spread-filter thresholds are considered:

- **Primary (05):** rel_spread $< 5\%$.
- **Relaxed (15):** rel_spread $< 15\%$.

The primary filter is used for all main results; the relaxed filter is used for the crisis-date robustness check (§7).

3.3 Maturity selection (Step 3)

Five target maturities (in calendar days) are selected from each chain: 7, 30, 93, 182, and 366 days. The actual expiry closest to each target is chosen.

3.4 Forward and discount factor calibration (Step 4)

For each selected maturity, the forward price F and discount factor $D = e^{-rT}$ are extracted by fitting put-call parity via OLS regression on paired strike prices. Concretely, writing $C_{\mathrm{mid}}(K) - P_{\mathrm{mid}}(K) = F\,D - D\,K$ gives the linear model

$$y_i = a - D\,K_i, \qquad y_i = C_{\mathrm{mid}}(K_i) - P_{\mathrm{mid}}(K_i),$$

from which $D = -\hat{\beta}_1$ and $F = \hat{\beta}_0/D$. This ensures the model forward price exactly matches the market-implied forward, a necessary condition for arbitrage-free pricing. Per-maturity effective rates are then

$$r_{\mathrm{eff}}(T) = -\ln D/T, \qquad q_{\mathrm{eff}}(T) = r_{\mathrm{eff}}(T) - \ln(F/S_0)/T,$$

where the spot proxy $S_0 = F_{\min}\,D_{\min}$ is computed from the shortest-dated maturity.

3.5 OTM selection and moneyness filter (Steps 5–6)

- **OTM rule:** retain puts for $K < F$, calls for $K \geq F$.
- **Moneyness filter:** retain strikes with $\kappa = K/F \in [0.85, 1.15]$.

The resulting quote counts per date and spread threshold are summarised in Table 1.

Table 1: OTM quote counts after all filters.

Date	Regime	Spread 5%	Spread 15%
2020-01-15	low-vol	224	—
2020-03-18	crisis	113	243
2021-09-15	recovery	575	—

The crisis surface under the 5% filter retains only 113 quotes across *two* maturities (30 d and 93 d); the 7 d, 182 d, and 366 d maturities are eliminated entirely by extremely wide bid-ask spreads characteristic of the COVID crash.

4 Pricing Methodology

4.1 B-Spline Galerkin inner pricer

European put prices are computed by solving the CGMY fractional backward FPDE on a log-price spatial domain $[x_{\min}, x_{\max}]$ with cubic ($p = 3$) cardinal B-spline basis functions and Crank–Nicolson (N_τ time steps) time-stepping. The spatial semi-discretisation produces the linear ODE system

$$M_h\,\dot{c}(\tau) + A_h\,c(\tau) = f(\tau),$$

where M_h is the B-spline mass matrix, A_h is the CGMY operator matrix (sum of a first-order drift term, a Toeplitz-structured fractional stiffness matrix, and a kill-rate term), and $c(\tau)$ is the coefficient vector. The CN scheme advances this system from $\tau = 0$ (expiry payoff) to $\tau = T$ (today).

4.2 Domain and grid

The default `make_grid` in the B-spline package centres the spatial domain at $\log K$ with width proportional to $1/G + 1/M$, which collapses to only ± 0.1 log-units for $G = M = 10$ — too narrow for active CGMY. For calibration a fixed-width grid is used instead:

$$[x_{\min}, x_{\max}] = [\log K_{\text{ref}} - 2,\ \log K_{\text{ref}} + 2],$$

covering $S \in [K_{\text{ref}}\, e^{-2},\ K_{\text{ref}}\, e^{2}]$, a ratio of approximately $1:55$ around the reference strike. Benchmark tests against the COS pricer confirm relative pricing errors below 0.25% at $N = 256$ under this domain choice.

4.3 Homogeneity pricing of a strike strip

Rather than solving one PDE per strike, the homogeneity identity

$$P(S_0, K; r, q, T) = \frac{K}{K_{\text{ref}}}\, P\left(S_0\, e^{-qT} \cdot \frac{K_{\text{ref}}}{K},\ K_{\text{ref}};\ r, q{=}0, T\right)$$

is used to price all strikes K_j in a single PDE solve with reference strike $K_{\text{ref}} = S_0$. The dividend yield q is absorbed into the effective spot $S_0^{\text{eff}} = S_0\, e^{-qT}$, which shifts the evaluation point inside the already-computed solution. The evaluation points are

$$x_j = \log S_0^{\text{eff}} + \log K_{\text{ref}} - \log K_j,$$

and put-call parity yields call prices when $K_j \geq F$. This reduces the per-maturity pricing cost from $O(n_K)$ PDE solves to a *single* solve plus n_K B-spline evaluations.

4.4 Caching of CGMY-independent matrices

Two matrices dominate the assembly cost:

- M_h (mass matrix): computed via `scipy.integrate.quad` numerical integration over B-spline products; cost $\approx 490\,\text{ms}$ at $N = 64$; depends only on $(N, p, h, x_{\min})$ — *independent of CGMY parameters.*
- D_h (derivative matrix): similarly $\approx 480\,\text{ms}$; also CGMY-independent.
- K_G, K_M (fractional stiffness matrices): computed via IFFT of tempered coefficients; cost $\approx 5\,\text{ms}$; *depends on* (G, M, Y).

The factory function `make_cached_pricer` pre-builds M_h and D_h *once* per maturity at calibration start, then returns a closure that only recomputes K_G, K_M per optimizer call. This yields a $\sim 100\times$ speedup: a full 5-maturity surface evaluation drops from $\approx 4.9\,\text{s}$ to $\approx 51\,\text{ms}$, making the two-phase calibration tractable within minutes.

5 Calibration Methodology

5.1 Parameter space and unconstrained transform

The physical parameter bounds are

$$C \in [10^{-3}, 20], \quad G \in [0.1, 80], \quad M \in [1.01, 80], \quad Y \in [1.05, 1.95].$$

To convert a constrained problem into an unconstrained one the bijective transform

$$\eta_1 = \log C, \quad \eta_2 = \log G, \quad \eta_3 = \log(M-1), \quad \eta_4 = \log\left(\frac{Y-1}{2-Y}\right) \tag{2}$$

is used, with inverse

$$C = e^{\eta_1}, \quad G = e^{\eta_2}, \quad M = 1 + e^{\eta_3}, \quad Y = \frac{2\,e^{\eta_4}+1}{1+e^{\eta_4}}.$$

The bounds on η are derived programmatically from the physical bounds (never guessed by hand), so any change to the physical bounds propagates automatically. The resulting η-space bounds are $\eta_1 \in [-6.91, 3.00]$, $\eta_2 \in [-2.30, 4.38]$, $\eta_3 \in [-4.61, 4.37]$, $\eta_4 \in [-2.94, 2.94]$.

5.2 Objective function

The calibration minimises the vega-normalised mean-squared price error

$$\mathcal{L}(\eta) = \frac{1}{N_{\text{quotes}}} \sum_{i=1}^{N_{\text{quotes}}} \left(\frac{V_{\text{model},i}(\eta) - V_{\text{mid},i}}{\max(\nu_i,\, \nu_{\text{floor}})} \right)^2, \tag{3}$$

where $V_{\text{model},i}$ is the model OTM price, $V_{\text{mid},i}$ is the market mid-price, ν_i is the Black–Scholes forward vega, and $\nu_{\text{floor}} = 10^{-3}$ prevents blow-up for deep OTM quotes with negligible sensitivity.

Dividing by vega converts the objective from a price-error MSE to an approximate IV-error MSE, so that near-ATM liquid quotes (high vega) and deep-OTM quotes (low vega) contribute roughly equally in implied-volatility units.

5.3 Two-phase optimisation

Phase 1: Differential Evolution (global search). Scipy's `differential_evolution` is run with a coarse $N = 32$, $N_\tau = 5$ cached pricer (sufficient for identifying the basin of attraction; $\approx 1.5\%$ price error relative to $N = 256$). Fixed settings (used for all dates, for reproducibility):

- `seed` $= 0$
- `maxiter` $= 300$ (early stopping: `tol` $= 10^{-7}$)
- `popsize` $= 15$ (population $= 15 \times 4 = 60$ individuals)
- `polish` $=$ `False`
- `mutation` $= (0.5,\ 1.5)$, `recombination` $= 0.7$

At ≈ 25 ms per evaluation the maximum budget of 18,060 function calls would take ≈ 7.5 minutes; in practice DE converged in 9 000–11 000 calls (≈ 4 minutes).

Phase 2: L-BFGS-B (local refinement). Starting from the DE solution, a high-accuracy $N = 64$, $N_\tau = 10$ cached pricer is substituted and `scipy.optimize.minimize` with `method='L-BFGS-B'` is run to convergence (`ftol` $= 10^{-12}$, `gtol` $= 10^{-8}$). At ≈ 51 ms per evaluation, 130–265 calls are required (10–14 s).

5.4 Diagnostics

After Phase 2, every OTM market quote is re-priced with the $N = 64$ pricer and the model price inverted to model implied volatility via Brent's method on the Black–1976 forward formula. The IV RMSE and maximum absolute IV error across all quotes are reported as calibration quality metrics.

6 Results

6.1 Calibrated parameters and fit quality

Table 2 reports the calibrated CGMY parameters, IV RMSE, and maximum IV error for the three primary (5%-spread) surfaces and the crisis robustness surface (15%-spread).

Table 2: Calibration results. Fixed optimizer settings: seed = 0, `de_maxiter` = 300, `de_popsize` = 15. IV errors in decimal volatility (multiply by 100 for vol-points).

Date	Regime	Spread	C^*	G^*	M^*	Y^*	RMSE (vp)	Max (vp)
2020-01-15	low-vol	5%	0.049	2.236	80.0	1.089	2.47	9.80
2020-03-18	crisis	5%	0.543	0.100	27.8	1.050	7.19	17.10
2020-03-18	crisis	15%	0.546	0.100	6.7	1.050	11.05	46.53
2021-09-15	recovery	5%	0.097	2.594	80.0	1.050	1.97	8.09

The non-crisis dates (2020-01-15 and 2021-09-15) achieve IV RMSEs of 2.47 and 1.97 vol-points respectively, comfortably within the 3 vp acceptance threshold. The crisis date under the primary 5% filter yields an RMSE of 7.19 vp — above threshold and discussed at length in §9.

6.2 Per-maturity breakdown

Table 3 disaggregates the IV RMSE by maturity for each primary (5%-spread) surface.

Table 3: Per-maturity IV RMSE (vol-points) for primary (5%) surfaces. "n" is the number of OTM quotes surviving all filters.

Maturity	T (y)	RMSE (vol-points) / n quotes		
		2020-01-15	2020-03-18 (05)	2021-09-15
7 d	0.0192	8.52 / 15	*filtered*	5.38 / 34
30 d	0.0822	1.63 / 61	14.10 / 19	1.46 / 266
93 d	0.2548	0.56 / 96	4.68 / 94	1.07 / 179
182 d	0.455–0.504	0.88 / 32	*filtered*	2.05 / 49
366 d	1.0027	1.76 / 19	*filtered*	2.44 / 42
All		**2.47 / 224**	**7.19 / 113**	**1.97 / 575**

A consistent pattern emerges across all successfully calibrated surfaces: the very short (7-day) maturity shows elevated errors (5–9 vp), while medium-term maturities (30–366 days) fit well (0.6–2.5 vp). This is a known limitation of CGMY and Lévy models more generally: discrete overnight jump risk and microstructure effects dominate very short-expiry pricing and are not captured by the continuous-time process.

6.3 Computational performance

Table 4 records the wall-clock time for each calibration run on a single CPU core.

The crisis 5%-surface run is notably faster (120 s) because only two maturities survived the spread filter, halving the per-evaluation cost of the cached pricer. For all five-maturity surfaces the total calibration time is 4.2–4.6 minutes.

Table 4: Computational timing per calibration run (single CPU core).

Date	Spread	n_{mats}	DE evals	L-BFGS-B evals	Total (s)
2020-01-15	5%	5	9,420	235	255
2020-03-18	5%	2	11,340	170	120
2020-03-18	15%	5	10,500	130	275
2021-09-15	5%	5	9,360	265	261

7 Crisis-Date Robustness Check

The data-roadmap (Section 5) flags the 2020-03-18 date as requiring a robustness test: does the calibrated parameter vector change materially when the spread filter is relaxed from 5% to 15%? Table 5 reports the absolute and relative differences.

Table 5: Crisis-date (2020-03-18) parameter robustness: spread05 vs spread15.

Parameter	Spread 5%	Spread 15%	$\lvert\Delta\rvert$	$\lvert\Delta\rvert/\lvert\cdot\rvert_{05}$
C^*	0.5431	0.5462	0.003	0.6%
G^*	0.100	0.100	0.000	0.0%
M^*	27.79	6.65	21.14	76.1%
Y^*	1.050	1.050	0.000	0.0%
IV RMSE	7.19 vp	11.05 vp	3.87 vp	53.8%

Finding. C^*, G^*, and Y^* are highly stable (changes $\leq 1\%$) under the relaxed spread filter. However, M^* shifts dramatically from 27.8 to 6.7 — a 76% reduction. This instability has a clear structural explanation: M controls the exponential decay rate of the right tail of the Lévy measure (upward jumps), and is therefore identified primarily by long-maturity OTM call options. The 5% filter retains *no* maturities beyond 93 d; the 15% filter recovers 5 maturities including 182 d and 366 d calls. With dramatically different surface shapes entering the objective, the optimizer finds a different M^* while holding the other three parameters near-constant.

The substantially worse RMSE under the 15% filter (11.05 vp vs 7.19 vp) indicates that the additional quotes admitted by the looser filter are noisy (wide spreads, stale quotes) and make the fitting problem harder, not easier. The 5% surface, though sparse, is more internally consistent.

For the paper's Section 7 the 5%-filter result is reported as the primary finding. The 15%-filter result is documented here as evidence that *crisis-date M^* should be interpreted with caution*: it is sensitive to which maturities survive the quality filter, and both estimates press against the feasible-set boundary (lower bound $G = 0.1$, lower bound $Y = 1.05$), suggesting the data do not uniquely identify the full CGMY parameter vector in this regime.

8 Financial Interpretation

8.1 Jump-intensity scaling with market regime

The most striking cross-date pattern is the C^* parameter:

$$C^*_{\text{low-vol}} = 0.049, \quad C^*_{\text{crisis}} = 0.543, \quad C^*_{\text{recovery}} = 0.097.$$

The crisis value is 5.6× **larger** than the recovery value and 11× **larger** than the low-vol value. Since C scales the overall intensity of the CGMY Lévy measure (more precisely, C multiplies the jump rate density at every size), a larger C^* means the risk-neutral measure assigns a substantially higher probability to large price moves. This is consistent with the known behaviour of risk-neutral CGMY parameters during stress periods: market participants price options as if the world is in a higher-activity jump regime, even after accounting for the elevated level of the IVS through the volatility-normalised objective.

8.2 Fine-structure index stability

The Y^* estimates are

$$Y^*_{\text{low-vol}} = 1.089, \quad Y^*_{\text{crisis}} = 1.050, \quad Y^*_{\text{recovery}} = 1.050.$$

The cross-regime range is 0.039 — a spread of only **3.6%** relative to the cross-date mean $\bar{Y} = 1.063$. This narrow range is consistent with the theoretical interpretation of Y as a structural fine-structure index governing the activity of small jumps, which changes slowly with market conditions and is largely regime-independent. All three estimates are in the infinite-variation regime $Y \in (1, 2)$, as expected for SPX options.

Boundary note. In two of the three dates ($Y^* = 1.050$) the optimizer converges to the lower bound of the Y search domain $[1.05, 1.95]$. This suggests the data would prefer Y slightly below 1.05. The bound is imposed by the theoretical requirement $Y > 1$ for infinite variation plus a numerical margin; the proximity to the boundary means Y^* should be interpreted as an upper-bound constraint rather than a well-identified interior minimum.

8.3 Tail-decay parameters

The G^* and M^* estimates reveal an interesting asymmetry:

- **Non-crisis dates:** $G^* \approx 2.2$–2.6, $M^* = 80$ (upper bound). A small G (slow left-tail decay) implies a heavier downside tail — consistent with the well-known negative skewness of SPX risk-neutral distributions. M^* hitting the upper bound means the optimizer cannot distinguish $M = 80$ from $M = \infty$: the data are insensitive to the right-tail decay rate, likely because the $\kappa \in [0.85, 1.15]$ moneyness window contains few deep OTM calls.
- **Crisis date (5% filter):** $G^* = 0.10$ (lower bound), $M^* = 27.8$. During the crisis *both* tails are affected, but the optimizer reaches the G lower bound, indicating a very heavy left tail that cannot be distinguished from an even heavier one within the search domain. M^* falls substantially (from 80 to 27.8), reflecting the market pricing some right-tail risk (large upward moves during crisis bear-market rallies).

9 Crisis-Date Fit Caveat

The 2020-03-18 calibration (5%-filter) yields an IV RMSE of **7.19 vol-points**, well above the 3 vp acceptance threshold met on the other two dates. This is not a failure of the calibration algorithm; it reflects two structural issues with the crisis surface:

(i) Sparse, two-maturity surface. The 5% spread filter retains only 113 quotes across the 30 d and 93 d maturities. Fitting a four-parameter model to a two-maturity surface is underdetermined: C and Y are primarily identified by the level and curvature of the smile, while G and M are primarily identified by the skew *across maturities* — information that requires at least three maturities.

(ii) Extremely elevated IV level. SPX ATM implied volatility on 2020-03-18 was approximately 80%. The worst-fit quotes are all near-ATM 30 d options (log-moneyness ≈ 0, IV ≈ 0.80), where the model underestimates IV by up to 17 vp. The CGMY process, calibrated to match the overall skew shape, cannot simultaneously match the extreme level without distorting the smile curvature. This is a known limitation of parametric Lévy models: they have limited flexibility to jointly match the level and curvature of the smile during crisis events.

(iii) Boundary convergence. The optimizer converges to $G^* = 0.10$ (lower bound) and $Y^* = 1.05$ (lower bound), indicating the loss surface does not have a well-defined interior minimum under these data. The reported parameter values are therefore constrained estimates rather than interior maximum-likelihood solutions.

Recommendation for Section 7 paper text. Report the crisis calibration result with the following caveat: "*The 2020-03-18 surface, reduced to two maturities by the 5% spread filter, yields an in-sample IV RMSE of 7.2 vol-points. We attribute this to the extreme market conditions during the COVID-19 crash rather than to a deficiency of the calibration procedure: the sparse surface is insufficient to identify all four CGMY parameters, and the model is asked to match implied volatilities near 80%, a regime outside the domain covered by typical CGMY calibrations. The crisis result is reported for completeness; the low-volatility and recovery dates, where the surface covers five maturities and the model is well-identified, are the primary evidence for the calibration methodology.*"

10 Grünwald–Letnikov Benchmark

To quantify the calibration-speed advantage of the B-Spline Galerkin scheme, we run an accuracy-matched speed comparison against a Grünwald–Letnikov (GL) finite-difference solver on the 2020-03-18 crisis surface.

10.1 GL solver

The GL scheme discretises the same backward FPDE on a uniform log-price grid using tempered Grünwald–Letnikov weights (the $p = 0$ version of the tempered symbol, without the B-spline hat function) and Crank–Nicolson time-stepping. The spatial accuracy is $O(h)$ compared to $O(h^{p-Y/2}) \approx O(h^{2.5})$ for the B-spline scheme at $p = 3$, $Y = 1.05$. The GL operator matrix is dense (lower + upper Toeplitz sum) and is assembled each call via vectorised `numpy` operations; LU factorisation uses `scipy.linalg.lu_factor` with no pre-build step.

10.2 Accuracy diagnostic

At the crisis calibrated parameters ($C = 0.543$, $G = 0.1$, $M = 27.8$, $Y = 1.05$), the GL pricer's maximum relative pricing error versus a high-resolution reference is:

Table 6: GL pricing accuracy at crisis parameters ($G = 0.1$, $Y = 1.05$).

N	Max. relative error
32	83.84%
64	5.94%
128	2.68%
256	1.74%

The large error at $N = 32$ arises because $G = 0.1$ implies nearly un-tempered left jumps that decay as $e^{-0.1\,kh}$ per lag k; at $h = 4/32 = 0.125$ the factor is 0.988 per step, so $N = 32$ grid points capture less than 34% of the total left-tail weight. At $N = 32$ the DE optimizer cannot distinguish the true minimum ($G = 0.1$, $Y = 1.05$) from a spurious flat region ($G = 80$, $Y = 1.95$). At $N = 64$ the error drops to 5.94%, sufficient for the optimizer to converge to the correct basin.

10.3 Accuracy-matched comparison

We compare the two methods at grid sizes chosen so that both achieve similar pricing accuracy on the crisis surface:

- **B-Spline:** $N_{\text{DE}} = 32$, $N_{\text{L-BFGS}} = 64$ (standard paper settings; $< 1\%$ error at $N = 32$).
- **GL:** $N_{\text{DE}} = 64$, $N_{\text{L-BFGS}} = 128$ (accuracy-matched; $\sim 6\%$ and $\sim 3\%$ error respectively).

All optimizer hyper-parameters are identical: `seed`=0, `de_maxiter`=300, `de_popsize`=15, `tol`=1e-7, `mutation`=(0.5, 1.5), `recombination`=0.7.

Table 7: Accuracy-matched GL benchmark on 2020-03-18 (crisis) surface. Optimizer settings identical for both methods.

Method	N_{DE}	$N_{\text{L-BFGS}}$	C^*	G^*	Y^*	IV RMSE	Total (s)
B-Spline Galerkin	32	64	0.543	0.100	1.050	7.19 vp	115
GL finite-diff.	64	128	0.582	0.100	1.050	6.66 vp	20

Both methods converge to the same economic region: $G^* = 0.100$, $Y^* = 1.050$ (exact agreement), C^* within 7.2%. The M^* parameter disagrees (27.8 vs. 43.4) but this is a known identification degeneracy: with only two maturities the crisis surface provides insufficient constraint on M (confirmed by the B-spline returning $M^* = 27.8$ at 5% spread vs. $M^* = 6.7$ at 15% spread on the same date; see Section 7).

10.4 Speed ratio

$$\alpha = \frac{T_{\text{GL}}}{T_{\text{B-Spline}}} = \frac{20\,\text{s}}{115\,\text{s}} = 0.17$$

GL calibration completes in **20 s** versus **115 s** for the B-spline, a **5.8× wall-time advantage for GL** at these accuracy levels.

Drivers of the GL speedup.

1. **Zero pre-build cost.** The B-spline requires a 5-second pre-build (two cached `scipy.quad` integrals per maturity for M_h, D_h), which amortises over many evaluations but adds fixed overhead. GL has no pre-build step.
2. **Cheaper per-evaluation.** GL at $N_{\text{DE}} = 64$ costs 1.4 ms per evaluation (two maturities) vs. 9.7 ms for the B-spline at $N_{\text{DE}} = 32$ — a 6.7× per-call advantage. The GL matrix build is $O(N^2)$ vectorised `numpy`; the B-spline involves IFFT calls and cached Toeplitz matrix–vector products.
3. **Grid-size ratio only 2×.** Because $O(h)$ convergence requires $N_{\text{GL}} = 2\,N_{\text{BS}}$ in this regime, and not a larger multiple, the 6.7× per-eval advantage is only partially offset.

B-Spline's accuracy advantage. The B-Spline scheme retains a decisive edge in pricing accuracy per grid point: $O(h^{2.5})$ vs. $O(h)$. For high-accuracy applications (e.g. pricing exotic products to 0.1 vp tolerance) the B-spline would require $N \approx 64$ where GL would need $N \geq 1024$, making GL's per-eval cost $(1024/64)^3 \approx 4096\times$ more expensive and its total calibration time orders of magnitude slower. The 5.8× GL advantage reported here applies specifically to the moderate-accuracy calibration regime ($\approx 5\,\text{vp}$ IV RMSE) and the particular cost structure of this implementation (scipy.quad-based B-spline pre-build vs. LAPACK-LU GL).

11 Conclusion

The B-Spline Galerkin + CN solver serves as an effective inner pricer for market calibration of the CGMY model to SPX option surfaces. The full pipeline — surface construction, forward/discount calibration, OTM selection, cached pricing, two-phase optimisation — runs in under 5 minutes per date on a single CPU core.

Key results:

- **Non-crisis dates:** IV RMSE of 1.97 vp (2021-09-15) and 2.47 vp (2020-01-15), comfortably within the 3 vp target.
- **Crisis date:** IV RMSE of 7.19 vp due to a sparse two-maturity surface and extreme IV levels; this is a data limitation, not an algorithmic failure.
- **Jump intensity** C^* is 5.6–11× elevated at the crisis date relative to the non-crisis dates, consistent with the risk-neutral measure implying a higher-activity jump regime during stress.
- **Fine-structure index** Y^* is stable across regimes (range 3.6% of mean), consistent with its theoretical interpretation as a structural parameter.
- **Crisis robustness:** C^*, G^*, Y^* are stable ($< 1\%$ change) under the 15%-spread filter; M^* shifts by 76%, reflecting the dependence of right-tail identification on long-maturity quote availability.

The cached-pricer architecture (pre-built M_h, D_h per maturity; IFFT-only rebuild per optimizer call) is the key computational innovation enabling this calibration throughput. Without caching, a single 5-maturity loss evaluation would require ≈ 5 s; with caching it requires ≈ 51 ms — a 100× speedup that transforms a 9-hour brute-force calibration into a 4-minute workflow.